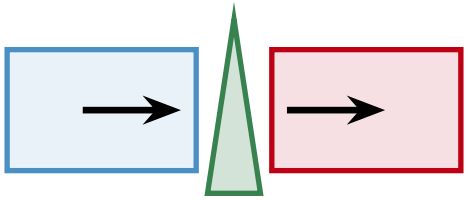
Chapter 8

The Costs of Taxation



A Tax Creates a Wedge

A per-unit tax separates the price buyers pay, P_B, from the amount sellers keep, P_S.

$$P_B - P_S = t.$$

Buyers decide how much to purchase using P_B. Sellers decide how much to supply using P_S.

The market clears when quantity demanded at P_B equals quantity supplied at P_S.

Three Groups in the Welfare Calculation

In a competitive market with no externalities, consumers receive consumer surplus (CS), and producers receive producer surplus (PS).

The government receives tax revenue, $R = tQ_T$, where Q_T is the quantity traded after the tax.

$$\text{Total surplus} = CS + PS + R.$$

Note that revenue is a transfer to the government. Counting it avoids treating every dollar paid in tax as a social loss.

Example: A Tax on Bubble Tea

Consider a hypothetical competitive market for bubble tea with no externalities.

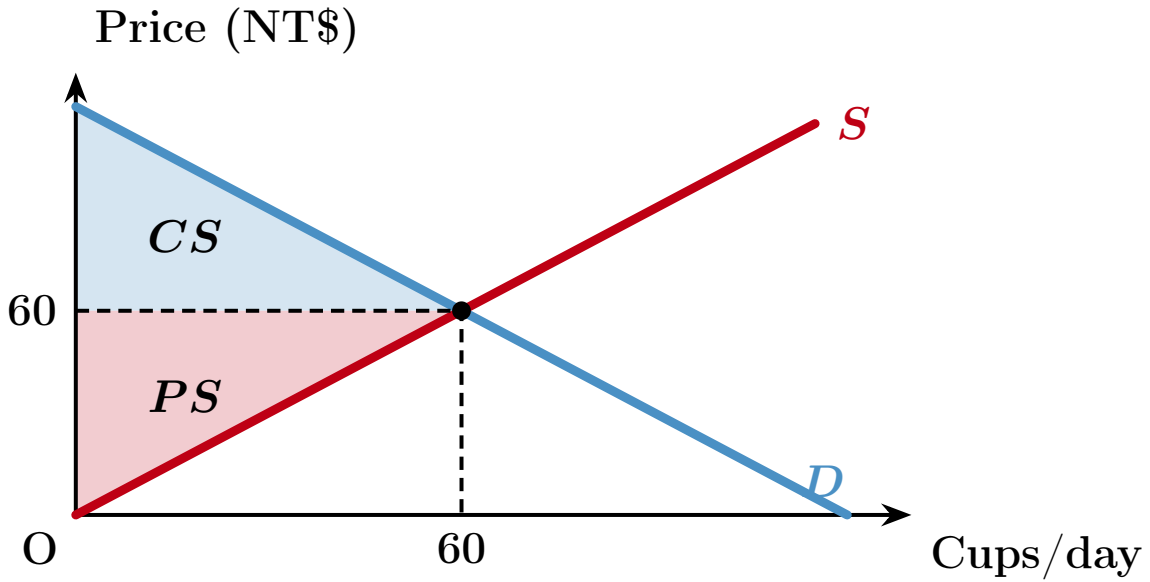
$$P_B = 120 - Q, \qquad P_S = Q.$$

Prices are NT$ per cup, and quantity is cups per day.

Without a tax, buyers and sellers face the same price. Thus, $120 - Q = Q$, giving $Q = 60$ and $P = 60$.

Surplus before the Tax

The blue and red triangles measure the gains to buyers and sellers.



$$CS = PS = \tfrac{1}{2}(60)(60) = 1{,}800, \qquad TS = 3{,}600.$$

Prices and Quantity with a Tax

Suppose the tax is NT$40 per cup. Substitute the inverse demand and supply curves into the tax wedge.

$$(120 - Q) - Q = 40.$$

Thus, $Q_T = 40$, $P_B = 80$, and $P_S = 40$.

The interpretation is that buyers pay NT$20 more per cup, and sellers keep NT$20 less. These are burdens per unit traded.

Who Sends the Tax Payment?

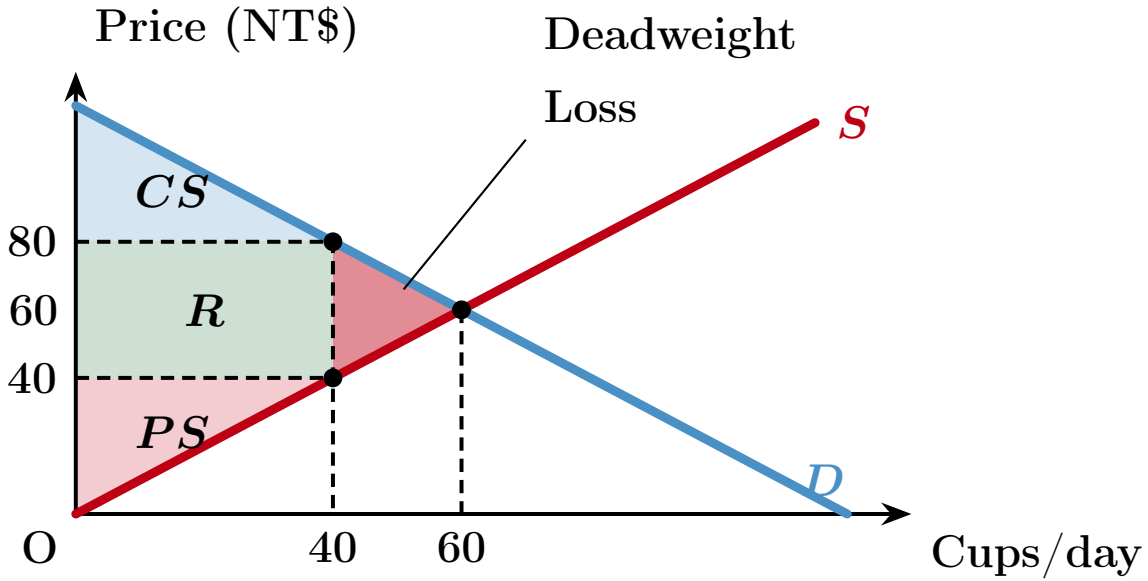
If sellers remit the NT$40 tax, buyers pay NT$80 to the shop. The shop sends NT$40 to the government and keeps NT$40.

If buyers remit the tax, they pay NT$40 to the shop and NT$40 to the government. Their total payment is still NT$80.

Either way, $P_B - P_S = 40$ and $Q_T = 40$. Who sends the payment does not determine the economic burden; demand and supply do. Surplus and revenue are unchanged.

Surplus and Revenue after the Tax

The entire green rectangle is revenue. The small triangle represents gains from trades that no longer occur.



$$CS = 800, \quad PS = 800, \quad R = 40(40) = 1{,}600.$$

Deadweight Loss

Definition. Deadweight loss is the reduction in total surplus caused by a market distortion.

In the bubble-tea market, total surplus after the tax is

$$800 + 800 + 1{,}600 = 3{,}200.$$

Thus, deadweight loss is $3{,}600 - 3{,}200 = \text{NT\$400}$ per day. Revenue itself is not deadweight loss.

Where the Lost Surplus Goes

Group	Before tax	After tax	Change
Consumers	1,800	800	−1,000
Producers	1,800	800	−1,000
Government	0	1,600	+1,600
Total	3,600	3,200	−400

All amounts are NT$ per day. Consumers and producers lose NT$2,000, but NT$1,600 becomes government revenue. The remaining NT$400 is lost surplus.

The Trades That Disappear

At 50 cups per day, willingness to pay for the marginal cup is NT$70 and its marginal cost is NT$50.

That cup would generate NT$20 of surplus. Yet no price can both cover the seller's cost and leave the buyer willing to purchase when the tax is NT$40.

Overall, the tax prevents trades between 40 and 60 cups even though willingness to pay exceeds marginal cost.

The Deadweight-Loss Triangle

For linear demand and supply with positive post-tax trade, lost surplus is a triangle with height t and base $Q_0 - Q_T$, where Q_0 is the no-tax quantity.

$$DWL = \tfrac{1}{2}t(Q_0 - Q_T).$$

For bubble tea,

$$DWL = \tfrac{1}{2}(40)(60 - 40) = 400.$$

Note that the formula measures lost gains from trade, not lost sales revenue.

Example: A Laundry Service

Mei values a monthly laundry service at NT$1,200. Lin's opportunity cost of providing it is NT$900. They agree on a price of NT$1,000.

Mei receives NT$200 in consumer surplus, and Lin receives NT$100 in producer surplus. Total surplus is NT$300.

Keep in mind that these hypothetical numbers describe one indivisible transaction.

A Tax Can Eliminate a Trade

A tax of NT$400 on the laundry transaction requires a buyer price at least $900 + 400 = 1{,}300$ for Lin to cover cost.

Mei will pay at most NT$1,200, so the transaction does not occur.

As a result, the government receives no revenue. Deadweight loss is the NT$300 of surplus that the transaction would have generated.

Elasticity Determines the Quantity Response

A tax causes a larger deadweight loss when it eliminates more mutually beneficial trades.

For linear curves, hold the initial equilibrium and the tax fixed. With the same upward-sloping supply, more elastic demand creates a larger quantity reduction.

Similarly, with the same downward-sloping demand, more elastic supply creates a larger quantity reduction. If either side is perfectly inelastic, quantity does not change and there are no lost trades.

Demand Responses with the Same Starting Point

Compare two demand curves passing through $P = 60$, $Q = 60$, with the same supply $P_S = Q$ and tax $t = 30$.

Demand	Quantity with tax	DWL
$P_B = 180 - 2Q$	50	150
$P_B = 90 - \frac{1}{2}Q$	40	300

The flatter demand curve has greater elasticity at the common starting point. It produces twice the quantity reduction and twice the deadweight loss.

Supply Responses with the Same Starting Point

Keep demand $P_B = 120 - Q$ and the tax $t = 30$. Both supply curves pass through $P = 60$, $Q = 60$.

Supply	Quantity with tax	DWL
$P_S = 2Q - 60$	50	150
$P_S = 30 + \frac{1}{2}Q$	40	300

The more elastic supply curve allows a larger reduction in production. Here too, the larger quantity response increases deadweight loss.

Perfectly Inelastic Demand

With upward-sloping supply and fixed quantity demanded, a tax does not change the quantity traded, provided supply can still meet that demand.

There are no lost trades, so deadweight loss is zero in this market model. Buyers bear the full tax through a higher price.

In other words, a large burden on buyers and a small deadweight loss can occur together.

Tax Incidence and Deadweight Loss

Incidence asks how the wedge is divided between buyers and sellers. The less elastic side bears more of the tax per unit.

Deadweight loss asks how much total surplus disappears. It depends on the reduction in quantity.

In summary, a group can bear most of the tax even when the deadweight loss is small.

Example: Choosing a Drink in Taipei

Suppose a hypothetical tax applies to only one bubble-tea chain in Taipei. Customers can switch to nearby untaxed shops. A tax on all bubble tea leaves fewer close untaxed substitutes, though customers can still choose other drinks.

This illustrates why demand for one brand is usually more elastic than demand for the whole category. Holding the initial equilibrium, responsive supply, and tax fixed, more elastic demand means more lost trades.

Adjustment also takes time. As customers change habits and firms change production or leave, demand and supply may become more elastic and tax distortions larger.

Labor Taxes and Work Decisions

A tax on labor creates a wedge between the cost of hiring a worker and the worker's take-home pay.

Responses include hours worked, a second earner taking a job, and retirement. One worker's unchanged hours can miss these responses.

A higher labor tax may also encourage a shift into **informal employment**, such as paid work whose earnings are not reported for tax purposes. For example, after a tax increase, a tutor switches from 10 reported hours to 10 unreported hours at the same pay.

In other words, moving work into informal employment can reduce taxable hours without reducing total hours worked. A smaller tax base alone does not measure lost surplus.

Larger Taxes in the Bubble-Tea Market

For the market $P_B = 120 - Q$ and $P_S = Q$, the tax wedge gives

$$Q_T = 60 - \frac{t}{2}, \qquad 0 \leq t \leq 120.$$

Therefore,

$$R = t\left(60 - \frac{t}{2}\right), \qquad DWL = \frac{t^2}{4}.$$

At the boundary $t = 120$, trade falls to zero. A higher tax also leaves no trade.

Tax Size, Revenue, and Lost Surplus

Tax per cup	Cups/day	Revenue	DWL
0	60	0	0
20	50	1,000	100
40	40	1,600	400
60	30	1,800	900
80	20	1,600	1,600
120	0	0	3,600

Revenue and deadweight loss are NT$ per day. Revenue first rises and then falls, while deadweight loss rises as trade shrinks.

Why Deadweight Loss Grows Quickly

With linear demand and supply and a positive quantity traded, doubling the tax doubles the quantity reduction.

The intuition is that both the height and base of the deadweight-loss triangle double, so its area becomes four times as large.

$$t : 20 \to 40 \quad \Rightarrow \quad DWL : 100 \to 400.$$

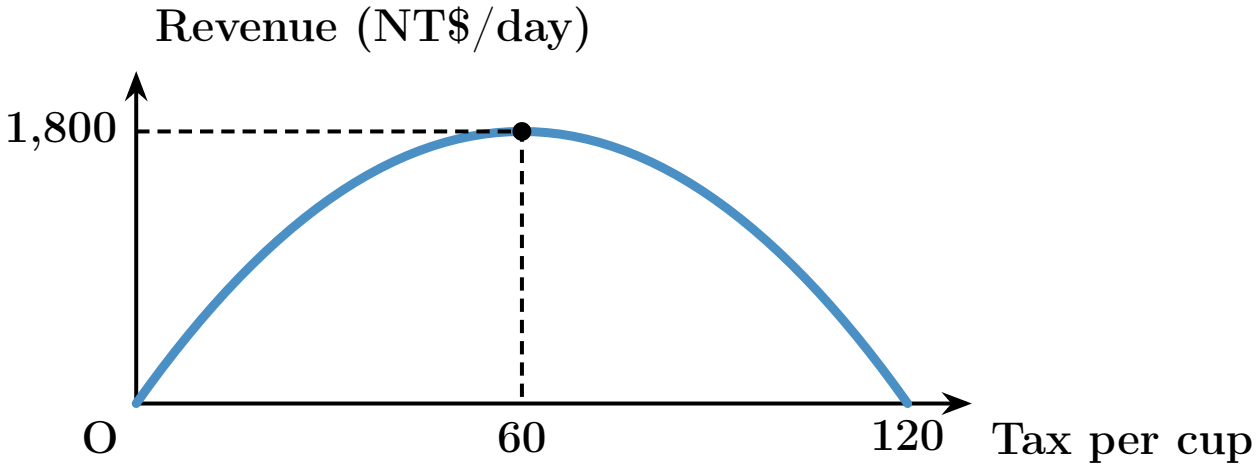
However, this proportionality stops once the tax eliminates all trade.

The Laffer Curve

Definition. The **Laffer curve** shows the relationship between the tax rate or tax amount and tax revenue.

For bubble tea, $Q_T = 60 - t/2$. For $0 \leq t \leq 120$,

$$R(t) = tQ_T = t\left(60 - \frac{t}{2}\right) = 60t - \frac{t^2}{2}.$$



Revenue peaks at $t = 60$: $R = 60(30) = \text{NT\$1,800}$ per day.

Does a Tax Cut Pay for Itself?

Reducing the bubble-tea tax from NT$80 to NT$60 raises revenue from NT$1,600 to NT$1,800.

By contrast, a cut from NT$40 to NT$20 lowers revenue from NT$1,600 to NT$1,000.

In summary, a tax cut raises revenue only when the tax-base expansion outweighs the lower tax on each unit. The existence of a Laffer curve does not establish where an actual economy lies on it.

Revenue Maximization Is Not the Policy Objective

The tax that collects the most revenue need not be the best tax.

In the bubble-tea example, a tax of NT$60 maximizes revenue but also destroys NT$900 of surplus per day.

Overall, evaluating a government program requires comparing its benefits with its resource costs and the distortions created by financing it. Distribution also matters for policy judgments.

Discussion

In the bubble-tea market, a tax rises from NT$20 to NT$40 per cup.

1. Does the tax burden per cup double for both buyers and sellers?
2. Does tax revenue double?
3. Does deadweight loss double?

Explain the different roles of the quantity traded and the quantity of trades lost.

Answer

Buyers' and sellers' burdens each rise from NT$10 to NT$20 per cup, so both double.

However, revenue rises from $20(50) = 1{,}000$ to $40(40) = 1{,}600$. It does not double because fewer cups are taxed.

Deadweight loss rises from $\frac{1}{2}(20)(10) = 100$ to $\frac{1}{2}(40)(20) = 400$. It quadruples because both the wedge and the lost quantity double.

Takeaways

In summary, taxes reduce surplus by preventing beneficial trades.

- Total surplus includes consumer surplus, producer surplus, and government revenue.
- More elastic supply or demand creates a larger deadweight loss, other things equal.
- Larger taxes can raise deadweight loss while reducing revenue.
- Tax policy involves efficiency, distribution, and the benefits of government spending.